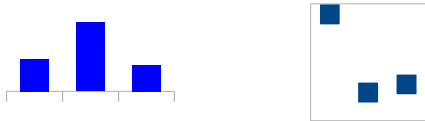


Prüfungsprotokoll - Data Literacy 18.02.2020
Exam Writeup - Data Literacy 18. Feb 2020

No utensils, devices or papers allowed in the exam. 90 mins of time, which was plenty (at least for me). Ms. Nieselt and Mr. Henning were giving the lecture to this.

1. Missing items in a dataset: name four methods to cope with missing data and describe one of them in greater detail.
2. In a longer text describing a medical case study, find the dependent and the independent variable. Explain, how bias can be unintentionally introduced in the study.
3. Name the „graphical or visual variables“ that are used to encode the values in these two plots, a bar chart and a scatterplot („graphical or visual variables“ probably referred to „channels“):



Describe the best and worst such channel for displaying ordinal data.

4. Explain the difference between classification, regression and ranking.
5. Which measurement unit do z-scores have? When are they useful? In which color scale would you visualize z-score values?
6. Find the odd ball:
You are given 12 balls, all equal in weight except for one that is either heavier or lighter than the others. You have a scale to measure the weights, it either shows equal weight or that one side is heavier (so, three possible outcomes). How often do you minimally need to use the scale to find the oddball? This was exactly the example from the lecture, so learn it by heart! Yet, they did not want you to write down the full procedure, only the information theory and Shannon part.
Shannon says: Don't come up with a complicated theory, just maximize the entropy!
The distribution over $12 \times 2 = 24$ possible outcomes is uniform
Thus, we need to collect $\log_2 24 = 4.58$ bits
Each question with three possible outcomes reveals at best $\log_2 3 = 1.59$ bits
We thus have to ask at least $\lceil 4.58 / 1.59 \rceil = \lceil 2.88 \rceil = 3$ questions
7. There are three classes: dog, mammal and pet. These intersections of the classes were given:
dog
mammal
dog-mammal
dog-mammal-pet
pet-mammal
Draw an UpSet that represents these intersections.
8. In another case study, there are eight dimensions recorded. How would you visualize this? How does your visualization react to new dimensions being added?
Either they referred to heatmaps or SPLOMs here.
9. Derive PCA: just learn the math slides for PCA and LDA by heart, for real. Mr. Henning wants to see this.
10. Three statements regarding PCA and LDA were given, it should be explained if they hold or not. For example one question was: does LDA use the full rank of the covariance matrix?
11. Ranking using topological ordering: how is it done?
Treating the rank ($A > B$) as an edge (A, B) yields a directed graph (Digraph).
The maximum acyclic subgraph problem seeks to find the smallest number of edges to delete such that a DAG (directed acyclic graph) is obtained. This is NP-complete.